

Communication

Date	15 March 2026
Team ID	TM-RISINGWATERS-04
Project Name	Rising Waters: AI Powered Flood Prediction System
Maximum Marks	1 Mark

Communication Plan:

Effective communication is essential for a successful project demonstration. This document outlines the communication strategy used within the team and with stakeholders throughout the project lifecycle, including how updates, issues, and feedback were managed.

S.No	Communication Type	Frequency	Channel / Tool	Participants	Purpose
1	Team Standup	Daily	Discord / Slack	All Core Developers	Discuss daily progress, align task priorities, and unblock code merges.
2	Progress Update	Weekly	GitHub Projects / Zoom	Full Team	Review sprint deliverables, inspect PyTorch training epochs, and update the Gantt chart.
3	Issue / Bug Discussion	As Needed	Slack & GitHub Issues	Affected Developers	Debug urgent issues (e.g., raster array out-of-memory or API failure).
4	Stakeholder Review	Bi-Weekly	Google Meet / Zoom	Team & Project Mentor	Present mid-sprint iterations and gather direct design feedback on mapping visualizations.
5	Final Demo Rehearsal	Once	Google Meet	Full Team	Perform end-to-end dry run of mock live telemetry triggers and dashboard charts.
6	Ad-hoc Peer Coding	As Needed	VS Code Live Share	Backend & ML Leads	Refining model weights, UNet tensor reshaping, and PostGIS optimization.

Communication Challenges & Resolutions:

S.No	Challenge Faced	Resolution / Action Taken
1	Asynchronous schedule alignment between ML training sprints and UI dashboard development.	Created a strict API contract using mock FastAPI endpoints early on, allowing frontend integration testing in parallel with active deep learning model tuning.
2	Inefficient manual hand-off of trained PyTorch model weights (.pth files) and spatial GeoJSON assets.	Set up an automated AWS S3 storage pipeline with webhook notifications configured on Discord, signaling immediate availability of fresh builds.
3	Ambiguous progress tracking across individual feature components during mid-development phase.	Implemented a Kanban workflow on GitHub Projects integrated with PR checks to strictly require automated test coverage before merging.